INTEL INTERNSHIP PROJECT PRESENTATION

TOPIC: AI-POWERED FORM FILLING ASSISTANT FOR INDIAN CITIZEN SERVICES

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~ Presented by GenAI-2

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<u>PROBLEM STATEMENT DETAILS</u>

Problem Statement - 3

AI-Powered Form Filling Assistant for Indian Citizen Services

Citizens often need to fill in multiple government service forms at Seva Kendras (e.g., for certificates, licenses, and welfare schemes). This project aims to build an AI tool that auto-fills these forms based on uploaded documents (Aadhaar, PAN, voter ID, etc.), reducing manual effort and errors.

Core Features:

- Upload documents (PDF/images) → Extract key details (name, DOB, address, ID numbers).
- Auto-map extracted data to the correct fields in government form templates.
- Allow user review and edit before submission.
- Support multiple Indian languages.
- Voice based input to fill the details

Proposed Architecture:

Document ingestion/Voice → OCR (for scanned docs) → Entity extraction → Form mapping → Output (filled form).

INTRODUCTION

Problem Definition: India, with a population of over 1.45 billion, still finds it difficult to provide truly accessible digital government services, even after major efforts under the Digital India initiative. Low digital and general literacy, clear gender gaps and huge language diversity, with 22 official languages and over 121 languages spoken by more than 10,000 people each, make government form-filling very challenging. For services like Aadhaar, PAN and voter registration, people often face long, complex procedures and forms mostly in English or Hindi, which leads to mistakes, rejections and reliance on cyber cafés or middlemen. These problems hit rural citizens, older people and persons with disabilities the hardest, showing the need for smart, multilingual and multimodal form-filling systems that can work with voice, text and documents in Indian languages to support truly inclusive service delivery with wide-scale adaptability.

Bridging these systemic and technical gaps demands a shift toward truly multilingual, inclusive systems that reflect India's linguistic diversity and uneven digital literacy. Such technical frameworks must enable citizens of all regions, languages, education levels and abilities to interact with government services on their own.

Proposed Solution: Hence, this project introduces **GovAI-Sub** (Governmental Form Submissions via AI) web-app, integrating the **IVAN AI** (Indic-Vision-ASR-NER AI), an end-to-end modular, lightweight, multilingual and multimodal pipeline for automating Indian government form filling. **IVAN AI** employs a dual-input architecture that supports document images or PDFs and voice-based input, ensuring accessibility for the elderly and users with motor, visual, vocal or auditory impairments.

TECHNICAL CHALLENGES AND SOLUTIONS

Technical Challenges	Proposed Mitigation Strategies (IVAN AI & GovAI-Sub)
English-centric systems with low Indic support and code-mixing	Multi-lingual system (OCR-ASR-Translation) supporting 17-22 Indic languages with code-mixing
VLMs fail on skewed fonts, changing layouts	Hybrid OCR Model (Paddle-Surya) immune to layouts and skewed-texts
High data scarcity, needing labor-intensive Indic or image data annotations	Transfer Learning and Domain-shift to English text-space via translation post-OCR, exploiting rich English-text datasets
Unimodal systems (Voice or Image)	Multimodal system (Voice and Image)
Cloud based systems incurring API costs, security threats and internet needs	Local, light-weight Models ready for edge devices and offline use
Sensitive to noise and Hallucinations	Highly immune to noise and hallucinations (OCR-ASR-Translation-NER)
Confined document types and Blackbox systems	Generalized across 9 document types with a scalable modular pipeline
Limited Adaptability and Accessibility	Wide-scale Adaptability and Accessibility

DATASETS

The various components of the project, developed in phases were trained and tested using real-life government document data and authentic speech/text samples to ensure practical accuracy and real-world adaptability.

Phase-I (Model-building and Prototyping) datasets:

- **OCR Dataset:** Document images with ground-truth text to evaluate OCR on complex layouts and scripts.
- **ASR Dataset:** Multilingual/code-mixed audio with ground-truth transcripts to test ASR on real-world audio.
- **Translation Dataset:** Multilingual/code-mixed sentences with source to target language labels and reference translations to evaluate the machine translation module for English–Indic and Indic–English directions.
- **IVAN-NER Dataset:** Manually annotated on limited, 100 text-samples derived and replicated from OCR-like noisy document texts, in CoNLL-style BIO format with 21 government-specific entity types.

Phase-II (Integration and Pipeline-Development) datasets:

- **OCR-NER-Pipeline-Dataset:** Document images with ground-truth texts, source-target language labels, translated English text and tagged entities to evaluate the OCR-NER pipeline.
- **ASR-Translator-Pipeline-Dataset:** Multilingual/code-mixed audio samples, ground-truth transcripts, source-target language labels and reference translations to evaluate the ASR-translator pipeline.

```
-DOCSTART- -X- O
Registration -X- _ O
No. -X- _ O
: -X- _ O
WB.BA.2013.40279 -X- _ O
Case -X- _ O
No. -X- _ O
: -X- _ O
9581 -X- _ O
Memo -X- _ O
No. -X- _ O
164 -X- _ O
Dated -X- _ O
12 -X- _ O
/ -X- _ O
10 -X- _ O
/ -X- _ O
2023 -X- _ O
Date -X- _ O
of -X- _ O
Registration -X- _ O
: -X- _ O
05 -X- _ O
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02 -X- _ O
/ -X- _ O
1956 -X- _ O
Name -X- _ O
of -X- _ O
the -X- _ O
Child -X- _ O
: -X- _ O
```

IVAN-NER Dataset

THE IVAN-OCR MODULE

Phase-I:

Surya-OCR was employed for text extraction from scanned documents, tested in three different variants, standard (detection→recognition), Hybrid Surya-LDR (layout analysis→detection→recognition) and Hybrid Surya-LR (layout alaysis→recognition) skipping the detection phase. The three variations signified proper text-detection for accurate results.

Phase-II:

From the observations of Phase-I, robust Text-Detection capable PaddleOCR was employed on Indian Document images, but extraction failed due to high computational needs. A hybrid combination was designed combining PaddleOCR’s robust, lightweight text-detector module and layout analysis module (in HPI modes) for accurate detection of skewed, small, non-linear texts (bounding-boxes) separately with Surya-OCR's robust multilingual Indic recognition module, following standard reading order for complex layouts, thus testing two hybrid variants, Hybrid DR OCR (paddle detection→surya recognition) and Hybrid LR OCR (paddle layout alaysis→surya recognition).

Final Observations and Conclusion: The Paddle-Surya Hybrid Engine of IVAN-OCR, can robustly extract code-mixed Indic text across 21 Indian languages from poor-quality, skewed and small-font documents without fine-tuning, while maintaining a lightweight approach. This approach is robust to diverse, non-standardized layouts and focuses on proper information retention in OCRed-text from documents rather than strict layout or reading-order modeling, unlike resource-intensive VLMs that often fail on noisy, low-quality or irregular government documents. But limitations exist on highly degraded documents and images.

THE IVAN-ASR MODULE

Phase-I:

For voice-interactions, IVAN-ASR module employed the OpenAI Whisper Large-v3 for multilingual speech transcription, supporting major Indic languages including English, Bengali, Hindi, Tamil, Telugu, Marathi and other code-mixed inputs. The model was configured in transcribe mode rather than translate mode to preserve linguistic nuances of Indic languages and code-mixed speech patterns prevalent in Indian contexts. The efficiency of ASR-based processing of user information containing multilingual alpha-numeric data was signified via this integration.

Final Observations and Conclusion: The Whisper model in IVAN-ASR demonstrates robust performance on multilingual and code-mixed alpha-numeric speech across 12 Indic-languages without requiring fine-tuning and exhibits considerable noise resilience. But accuracy degrades under high background noise conditions and with highly complex accents or vocal variations, occasionally resulting in transcription errors.

THE IVAN-TRANSLATOR MODULE

Phase-I:

The system employed a customized approach combining PolyGlot for rapid source language detection coupled with Facebook's nllb-200-distilled-600M model for translation, facilitating conversion to an English-centric text space post-OCR and translated information generation for direct multilingual form-filling post-ASR, while maintaining computational efficiency. PolyGlot identifies single or multiple source languages from noisy mixed-language inputs, with detected language information forwarded alongside the source text and target language specification to the NLLB model for translation execution. The distilled NLLB-200 variant significantly reduced computational overhead and eased the dataset limitations in this domain, making it suitable for frequent fallback operations within the IVAN processing pipeline.

Final Observations and Conclusion: The customized module design demonstrates robust handling of code-mixed inputs from both OCR and ASR pathways through normalization into a unified target language representation, while being extremely light-weight and effectively managing noise, spelling errors and occasional model hallucinations. The PolyGlot and NLLB-200 models supports 11 and 22 Indic languages respectively. However, the module exhibits limitations in translating numeric strings and demonstrates reduced accuracy for languages with insufficient training representation in the NLLB-200 corpus, resulting in translation failures for low-resource language pairs.

THE IVAN-NER MODULE

Phase-I:

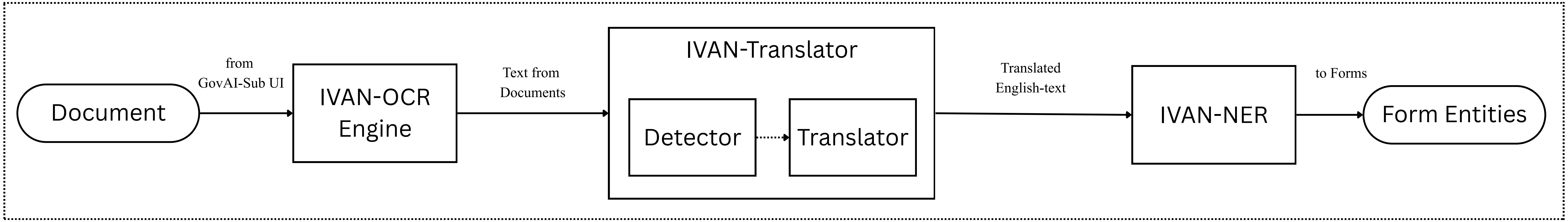
The system implemented a domain-adapted version of BERT-Large-NER model fine-tuned on the IVAN-NER Dataset to generate the BERT-Doc-NER as the IVAN-NER Module, tailored for information extraction from OCR-generated government document texts. The model was trained and validated using an 80:20 split of the 100 samples in IVAN-NER Dataset. By translating multilingual OCR output into an unified English text space before NER, the system simplifies training, reduces manual image-level or Indic scripts annotation and thus improves support for diverse documents.

Phase-II:

In this phase, the BERT-Doc-NER-Max was formulated, by fine-tuning BERT-Doc-NER on the complete dataset of 100 samples instead of the 80:20 split as of Phase-I to enhance pattern generalization and model performance improvements.

Final Observations and Conclusion: The two-stage training strategy employed for IVAN-NER involving initial training-validation on an 80:20 split followed by fine-tuning on the complete 100-sample dataset revealed that BERT-Doc-NER-Max outperformed BERT-Doc-NER, suggesting that increased training data would substantially enhance pattern generalization despite current limitations of IVAN-NER attributed to the small dataset size due to the legal constraints of unauthorized replication data from government documents. But, the novel methodology of leveraging resource-efficient, easily replicable and annotatable English-text datasets, significantly addresses the complex data preparation challenges prevalent in current research, offering a scalable approach for future improvements in training data-augmentation.

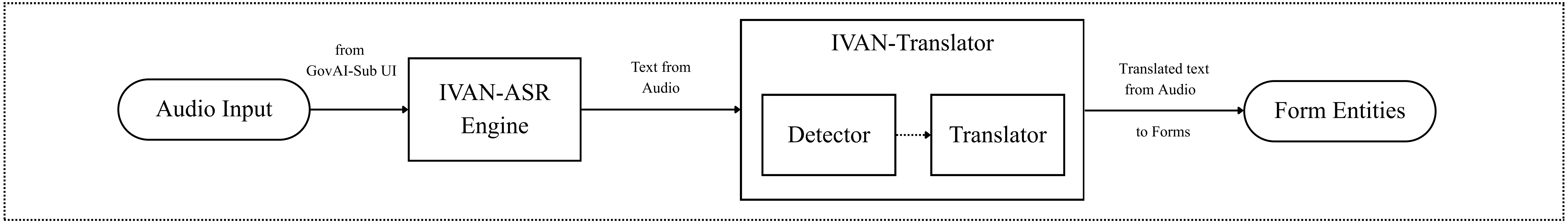
OCR-TRANSLATION-NER PIPELINE (PHASE-II)



This pipeline automates form-filling tasks by processing uploaded document image or PDF and extracting key information from them for filling forms.

- **Stage 1:** The uploaded document image or PDF is processed by IVAN-OCR to extract raw text blocks.
- **Stage 2:** IVAN-Translator auto-detects the source language(s) from the OCR-text and translates them to English.
- **Stage 3:** The English texts from Stage 2 is fed into the BERT-Doc-NER-Max (current IVAN-NER model) to extract the relevant target entities for the process, in a JSON format suitable for filling in forms.
- **Stage 4:** The form specific entities derived in Stage 3 as JSON are re-translated by the IVAN-Translator from English to the language in which the user is filling the form in and is directly mapped to form-fields.

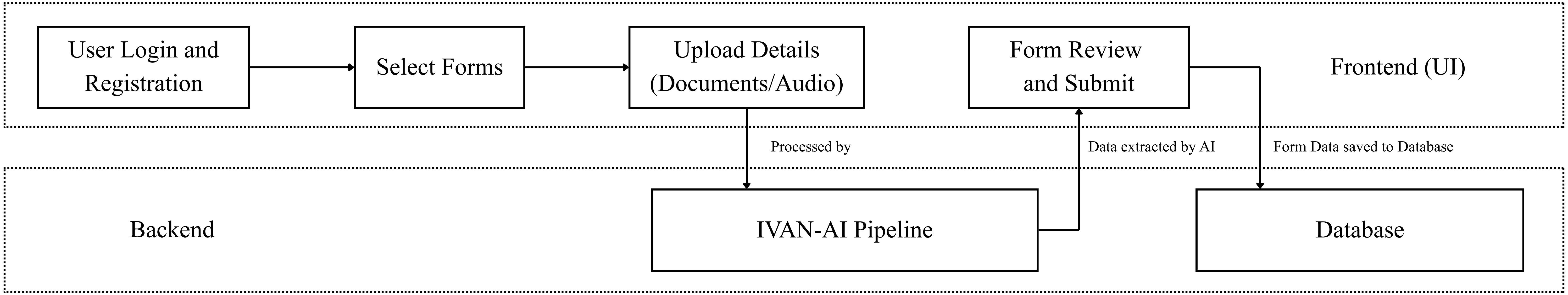
ASR-TRANSLATION PIPELINE (PHASE-II)



Designed for accessibility, this pipeline automates form-filling tasks by processing user information as input speech. By soliciting direct field-specific information from the user, this pipeline bypasses the NER stage reducing complexity (e.g., "Speak your Aadhaar number").

- **Stage 1:** Real-time audio information is captured and transcribed by IVAN-ASR into raw text as JSON.
- **Stage 2:** The transcribed ASR-text in JSON format (potentially code-mixed) is translated by IVAN-Translator into the target language of the form, producing a translated JSON output.
- **Stage 3:** The translated information from Stage 2 is directly mapped to the corresponding form fields.

GOVAI-SUB WEB APP (PHASE-II)



The GovAI-Sub web application workflow comprises the following steps,

- Users register with name, email, phone number and password for secure account creation, or log in if already registered.
- Upon login, users select their desired form and preferred language from supported options (English, Hindi, Bengali, Marathi, Tamil and Telugu), to load the forms directly in the selected language.
- Users are given choice for document upload or voice-based input, triggering the specific pipeline to automate form-filling.

The GovAI-Sub web application provides an intuitive user interface for automated form filling, catering to diverse user groups across varying age demographics, literacy levels, language proficiencies and disability profiles, thereby significantly enhancing accessibility and system adaptability for inclusive government service delivery.

GOVAI-SUB WEB APP INTERFACE DESIGN

Fig1: Home Page

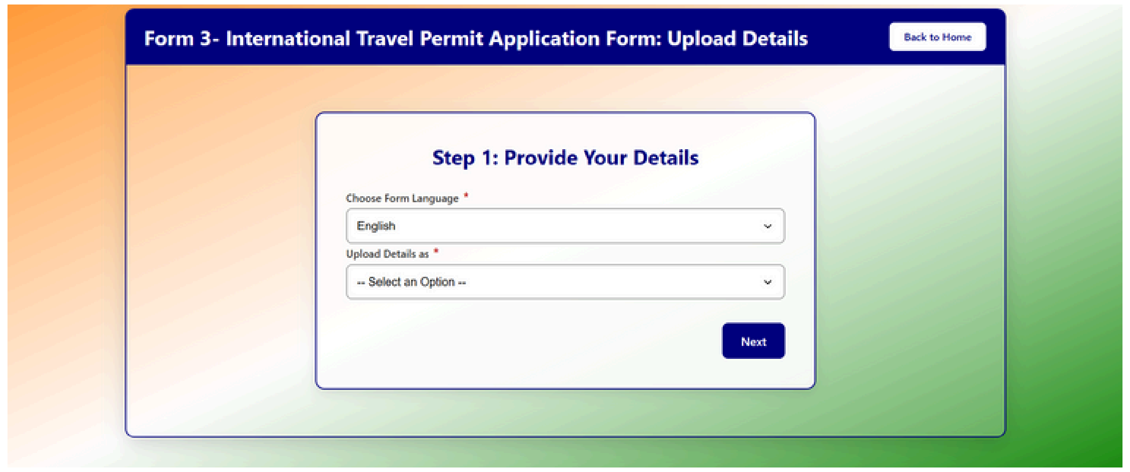


Fig2: Language Selection and Data upload Page (currently inactive)

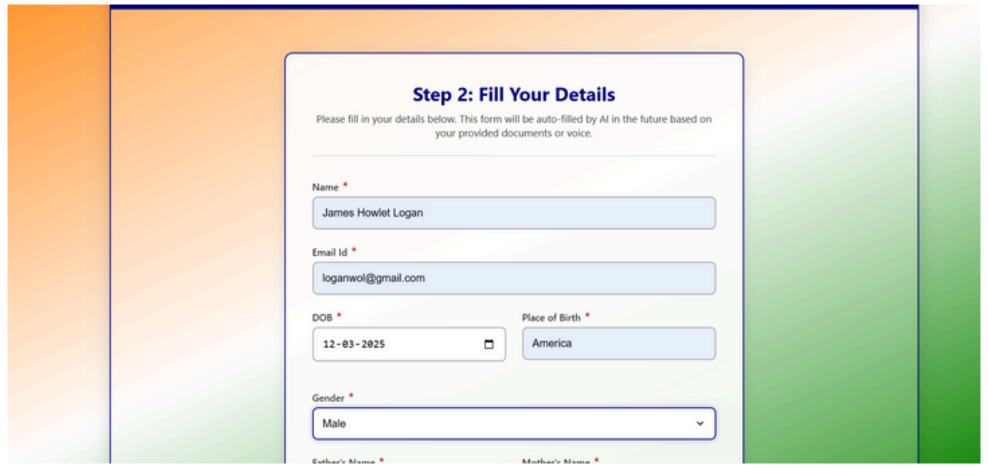


Fig3: Form Review and Submission Page (the auto-filling happens here)

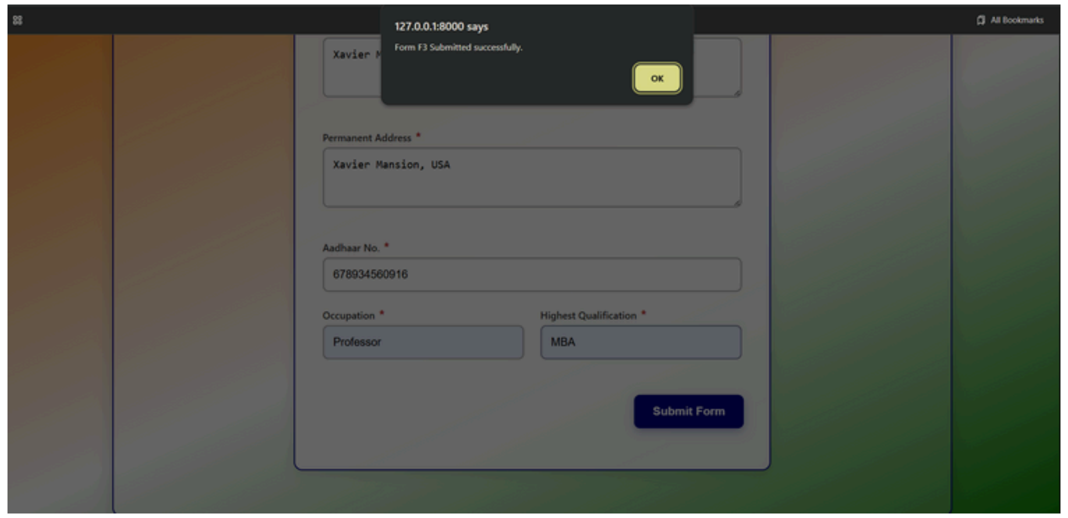


Fig4: Easy Form Submissions

Fig5: User Profile Page to verify saved user data

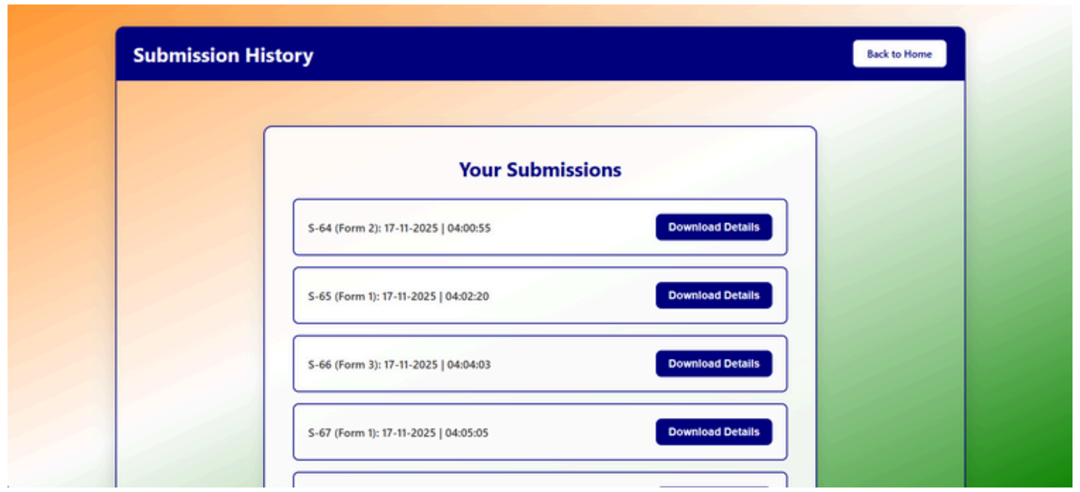


Fig6: Submission History for Submission Tracking and Download Details option to download filled Form for further processing

BACKEND DATABASE DESIGN

Table 1: Users

- user_id (PK)
- name
- phone_no (AK, Unique)
- email_id (AK, Unique)
- password (Hashed)
- dob
- is_active (Binary)

Table 2: UserAddress

- address_id (PK)
- session_id (FK)
- user_id (FK)
- current_address
- permanent_address

Table 3: Forms

- form_id (PK)
- name (AK, Unique)
- languages
- description
- proformas
- category

Table 7: PersonalInfo

- info_id (PK)
- session_id (FK)
- user_id (FK)
- birth_place
- gender
- aadhaar_no
- pan_no
- father
- mother
- spouse
- qualification
- occupation

Table 4: Sessions

- session_id (PK)
- user_id (FK)
- form_id (FK)
- form_language
- submission_date_time

Table 5: BankDetails

- user_id (PK)
- name
- phone_no (AK, Unique)
- email_id (AK, Unique)
- password (Hashed)
- dob
- is_active (Binary)

Table 6: PensionDetails

- user_id (PK)
- name
- phone_no (AK, Unique)
- email_id (AK, Unique)
- password (Hashed)
- dob
- is_active (Binary)

GovAI-Sub Web App Session-Management Logic:

- if users.is_active = 0 and (sessions.form_id = null and sessions.submission_date_time = null), then
  → no form filling, system in stable state (no Rollback needed)
- if users.is_active = 0 and (sessions.form_id = not null and sessions.submission_date_time = null), then
  → abrupt form closing, system in unstable state (Rollback needed)

RESULTS (PHASE-I)

Models	Accuracy Score (%)	Evaluation Metric	time(s)
Surya OCR	56.8	Normalized Levenshtein Dist. (Acc.)	5.43
Surya LDR	41.43	Normalized Levenshtein Dist. (Acc.)	30.9
Surya LR	59.02	Normalized Levenshtein Dist. (Acc.)	17.3
IVAN-ASR	73.7	Normalized Levenshtein Dist. (Acc.)	3.02
IVAN-Translator	47.4	Mean Aggregate of Detection (custom Penalty-based) & Translation (Normalized Levenshtein Dist. (Acc.))	0.297
IVAN-NER	73.49	F1 Score	0.062

RESULTS (PHASE-II)

Models	f1-score	det f1_score	precision	recall	overall score	time(s)
Hybrid LR OCR	**62.24**	-	-	-	-	8.03
Hybrid DR OCR	**77.81**	-	-	-	-	3.46
IVAN-ASR	**92.33**	-	91.83	92.87	-	3.02
IVAN-Translator	94.78	78.33	94.63	94.94	**86.55**	0.297
IVAN-NER	**79.8**	-	-	-	-	0.062

Evaluation Metrics Used: Considered "bag-of-words" based F1 Score for evaluating OCR model, BERTScore for evaluating ASR and Translator models, "bag-of-words" based F1 Score for evaluating language auto-detection model and simple F1 Score for evaluating NER model.

PIPELINE TESTING, LIMITATIONS AND CONCLUSIONS

Pipeines	Overall score (%)	time(s)
OCR-Translator-NER	81.46	6.33
ASR-Translator	84.35	3.027

- The pipeline testing was done on the improvements made in Phase-II of the project, with IVAN-AI (Hybrid DR OCR), IVAN-Translator (PolyGlot + NLLB), IVAN-ASR (OpenAI Whisper) and IVAN-NER (BERT-Doc-NER-Max)
- Though integration of IVAN-AI with GovAI-Sub web app was not possible due to localized GPU unavailability, but the IVAN AI module and GovAI-Sub module were tested individually, mimicking a real world deployment scenario.
- Pipeline efficiency is degraded by multiple factors including ASR noise sensitivity and transcription errors, outdated PolyGlot library, inadequate training of nllb-200-distilled-600M model resulting in failures on numeric strings and low-resource languages, OCR errors from highly degraded images and most critically, insufficient training data for the NER module (limited set of manually annotated 100 samples), considering legal constraints of replicating government data.

Conclusion: Despite these challenges, the IVAN-AI and GovAI-Sub system provides a robust, end-to-end, modular pipeline integrated successfully for automating Indic form-filing tasks with considerable benchmarks and novelties for being developed on such a complex, resource-constrained NLP domain. It counters significant research gaps, with high enhancement potential.

TOOLS USED AND FUTURE DIRECTIONS

Tools and Models used: Surya OCR and PaddleOCR for document text extraction, PolyGlot for language auto-detection, NLLB-200-distilled-600M for machine translation, BERT-large-NER for entity extraction, OpenAI Whisper for ASR, Label Studio facilitates dataset annotation for model training and Python3 for model coding, training, integration and testing.

Future Directions:

- Data Augmentation with noisy samples to expand IVAN-NER Dataset (1000+ samples approx.), with legal permissions for the same to replicate government document data into a dataset.
- Integrate IVAN AI module into the GovAI-Sub web app to generate a finished Interface for actual form-filling experiences.
- Replace outdated PolyGlot module with a modern state-of-the-art language detection module.
- Mitigate translation and ASR errors to transcriptions, noise, ascent, numeric-strings, etc., with simple spell-checker and rule based approaches for simplicity or advanced AIML based approaches.
- Improve OCR performance with explicit image preprocessing steps, that are absent in current IVAN-OCR framework.
- Use better evaluation strategy for OCR testing focusing solely on Information Retrieval principles, on the degree of necessary information retained by OCR, rather than penalizing the module for basic document texts in output that do not cater to important information in for form-filling.
- Design security, privacy and edge deployment principles and strategies for IVAN AI.

THANK YOU